Av = 21

28/40

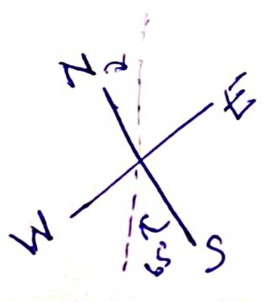
BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
First Semester 2024-25
CE F213: Surveying (Closed book)
Instructor-In-Charge: Prof. S.N. Patel

Name: MANAS CHOUDHARY **Duration:** 50 minutes

ID No: 2023A2PS1342P **Max Marks:** 40

Instructions:
1. Each question carries **ONE** mark 2. No negative marking 3. Calculators are allowed

1. Magnetic declination = Magnetic meridian – True meridian
   a. True b. False
2. In whole circle bearing, the bearing of a line is represented as C
   a. Back bearing = Fore bearing + 180^0 c. Back bearing = Fore bearing $\pm 180^0$
   b. Back bearing = Fore bearing + 180^0 d. None of the above
3. In a prismatic compass, the magnetic needle is of edge bar type
   a. True b. False
4. In a prismatic compass, the graduated ring is attached to the needle and does not rotate with line of sight
   a. True b. False
5. Convert the following quadrantal bearing into whole circle bearings,
   a. $N35^0 40'E$ = 35° 40'
   b. $S65^0 20'W$ = 245° 20'
6. The fore bearing of lines AB and BC are $135^0 40'$ and $45^0 20'$, respectively. The included angle B = 89° 40' 0''
7. Choose the correct statement(s).
   i. With a prismatic compass, sighting and reading can be done simultaneously.
   ii. Levelling of the compass is achieved by eye judgment.
   a. Only (i) is correct c. Both (i) and (ii) are correct
   b. Only (ii) is correct d. Both (i) and (ii) are wrong
8. The sum of internal interior angles of a closed traverse having n sides $(2n-4) \times 90°$
9. Which of the following is not a temporary adjustment of the prismatic compass?
   a. Centring c. Focusing prism
   b. Levelling d. Adjusting sight vane
10. Plumb bob is hung from the centre of the compass to ensure exact centring.
11. The vertical distance between two contour lines is called C
    a. Contour gradient c. Contour interval
    b. Vertical equivalent d. Horizontal line



45° 20' – 315° 40' + 360
44° 20

Not Answered

12. The line formed along the intersection of the ground surface and a level surface is known as

a. Contour line
b. Watershed line
c. Level line
d. Saddle

13. A contour line intersects a ridge line or valley line in a pattern, which can be exactly depicted as

a. Oblique
b. Vertical
c. Perpendicular
d. None of the above

14. The zone at which contour lines touch one another is indicated as

a. Level surface
b. Vertical cliff
c. Overhanging cliff
d. Horizontal surface

15. The contour depicting higher values inside the loop depicts

a. Level ground
b. Hilly areas
c. Depression
d. Both a and b

16. Locate the contours having reduced levels 53 m and 54 m between points A and B. The reduced levels of point A and B are 52.60 m and 55.80 m, respectively. The distance of point B from point A is 16 m.

a. 2 m and 9 m from A
b. 2 m and 7 m from A
c. 1 m and 3 m from A
d. 4 m and 11 m from A

17. The contour interval on a map is 13 m. If the upward gradient of 1 in 40 is required to be drawn between two points, what will be the horizontal equivalent?

a. 240 m
b. 520 m
c. 780 m
d. None of the above

18. Highway alignments are typically taken along

a. Valley line
b. Ridge line
c. Across the counter line
d. All the above

19. Normally contour lines of different elevations cannot cross each other ________

a. True
b. False

20. A series of closed contours on a map indicates:

a. Closed traverse
b. Summit
c. Depression
d. Both b and c

21. A levelling staff is a straight rectangular rod with graduations. The foot of the staff represents zero reading, and it is kept vertical at the point where the measurement is taken. From the level, a horizontal sight is taken. Which of the following is correct?

a. The staff is placed horizontally and sighted vertically
b. The foot of the staff represents the highest reading.
c. The staff is placed at a slant and sighted diagonally
d. The staff is placed vertically and slighted horizontally

22. What is the primary purpose of the pointed metal shoes at the ends of a tripod stands legs?

a. To prevent the legs from bending under heavy loads.
c. To increase the height of the tripod stand

52.60
A

B 55.80

62 400 125

b. To provide a good grip on the surface for stability

d. To allow the tripod to rotate smoothly on the ground

23. What is the configuration of the upper part of a telescopic staff?
    a. The upper part is hollow from inside
    b. The upper part is solid
    c. The upper part is flexible
    d. The upper part is adjustable by screws
24. How does the telescopic staff achieve its total length of 4m or 5m?
    a. The central part can slide into the upper part, and the upper part into the lower part
    b. The upper part slides into the central part, and the central part into the lower part
    c. The lower part slides into the upper part, and the upper part into the central part
    d. The central part is fixed, and the upper part extends outward
25. What material are cross hairs in surveying instruments typically made of?
    a. Copper wire
    b. Nylon thread
    c. Spider web, platinum filament, or silk
    d. Aluminium or glass fibers
26. What is the term used when the image of an object does not fall exactly on the plane of cross hairs, making accurate sighting impossible?
    a. Diffraction
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27. What is the term used for the last reading taken from the instrument before shifting it?
    a. Backsight
    b. Intermediate sight
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    d. Level sight
28. Which reading is always taken first after setting the instrument?
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29. If the height of the instrument (HI) is 101.01 m and the foresight (FS) is 2.42 m, what will be the reduced level (RL)?
    a. 98.59 m
    b. 100.12 m
    c. 101.85 m
    d. None
30. Which of the following is a correct relationship in leveling calculations?
    a. $\sum BS - \sum FS =$ Last RL - First RL
    b. $\sum FS = \sum IS$
    c. $\sum BS = \sum$Rise - Fall
    d. None
31. A check line in a chain surveying?
    a. Checks the accuracy of the framework
    b. Enables the surveyor to locate the interior details which are far away from the main chain lines.
    c. Fixes up the directions of all other lines
    d. all of the above
32. The length of a line measured with a 20 m chain was found to be 250 m. Calculate the true length of the line if the chain was 10 cm too long.
    a. 252.25 m
    b. 251.25 m
    c. 225.25 m
    d. 221.25 m

33. The method of measuring distance by pacing is chiefly used in?

a. Reconnaissance surveys
b. Preliminary surveys
c. Location surveys
d. All of these

34. Direct ranging is possible only when the end stations are?

a. Close to each other
b. Not more than 100 m apart
c. Mutually intervisible
d. Located at highest points in the sea

35. When the measured length is less than the actual length, the error is known as?

a. Positive error
b. Negative error
c. Compensating error
d. Instrumental error

36. The tension, at which the effects of pull and sag for a tape are neutralised, is known as?

a. Initial tension
b. Absolute tension
c. Surface tension
d. Normal tension

37. A plumb bob is required?

a. When measuring distances along slopes in a hilly country
b. For accurate centring of a theodolite over a station mark
c. For testing the verticality of ranging poles
d. All of the above

38. When the length of chain used in measuring distance is longer than the standard length, the error in measured distance will be?

a. Positive error
b. Negative error
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d. None of the above

39. A surveyor measured the distance between two points on the plan taking to a scale of 1cm = 40 m and the result was 468 m. But actual scale is 1 cm = 20 m. Find the actual distance between the two points.

a. 992 m
b. 936 m
c. 987 m
d. 234 m

40. The accuracy in laying down the perpendicular offsets and in measuring them depends upon?

a. Scale of plotting
b. Length of offset
c. Importance of the object
d. All of these

MASTER

Answers Labquiz-1.

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